

| Course Code | Course Name                                  | Theory                         |        |                        |              |                        | Term work | Pract / Oral | Total |
|-------------|----------------------------------------------|--------------------------------|--------|------------------------|--------------|------------------------|-----------|--------------|-------|
|             |                                              | Internal Assessment Test (IAT) |        |                        | End Sem Exam | Exam Duration (in Hrs) |           |              |       |
|             |                                              | IAT-I                          | IAT-II | IAT-I + IAT-II (Total) |              |                        |           |              |       |
| ESC102      | Basic Electrical and Electronics Engineering | 20                             | 20     | 40                     | 60           | 2                      | -         | -            | 100   |

### Course Objectives:

1. To provide knowledge on fundamentals of DC circuits
2. To provide knowledge of single phase and three phase AC circuits.
3. To inculcate fundamental knowledge of 1- $\Phi$  transformer.
4. To provide basic knowledge on fundamentals of DC and AC machines.
5. To provide knowledge of special purpose Diodes.
6. To provide knowledge of Transistor.

### Course Outcomes:

- 1) Apply various network theorems to determine the circuit response / behavior.
- 2) Evaluate and analyze 1- $\Phi$  and 3- $\Phi$  AC circuits.
- 3) Understand the construction, operation and applications of 1- $\Phi$  transformers.
- 4) Illustrate the working principle of 3- $\Phi$ , 1- $\Phi$  Induction motors and DC Motors.
- 5) Study the construction, operation and applications of some special purpose Diodes.
- 6) Study construction, operation and applications of some Transistors.

### DETAILED SYLLABUS:

| Sr. No. | Name of Module | Detailed Content                                                                                                                                                                                                                                                                                                                  | Hours | CO Mapping |
|---------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|
| 0       | Prerequisite   | Resistance, inductance, capacitance, series and parallel connections of resistance, concepts of voltage, current, power and energy and its units. Magnetic circuits, MMF, Magnetic field strength, reluctance.                                                                                                                    |       |            |
| I       | 01             | DC Circuits: (Only independent sources) Kirchhoff's Laws, Ideal and Practical Voltage and Current Sources, Source Transformation, Mesh and Nodal Analysis (no super node and super mesh) Star-Delta / Delta-Star Transformations, Superposition Theorem, Thevenin's Theorem, Norton's Theorem and Maximum Power Transfer Theorem. | 10    | CO1        |
| II      | 02             | AC Circuits: Generation of alternating voltage, basic definitions, average and RMS values, phasor and phase difference, sums on phasors, Single-phase ac series                                                                                                                                                                   | 12    | CO2        |

|     |    |                                                                                                                                                                                                                                                                                         |    |     |
|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
|     |    | and parallel circuits consisting of R, L, C, RL, RC, RLC combinations, definitions - real, reactive and apparent power, admittance (Y), Series and parallel resonance (only theory). Generation of Three-Phase Voltages, voltage & current relationships in Star and Delta Connections. |    |     |
| III | 03 | Single Phase Transformer: (Numerical are not expected) Working principle of single-phase transformer, types of single-phase transformer, transformation ratio, actual (practical) and ideal transformer, Transformer losses, efficiency, applications of transformer.                   | 04 | CO3 |
| IV  | 04 | Electrical Machines: (Numerical are not expected) principle of operation, constructional details, classification and applications of DC Motor, three-phase induction motor, Single-Phase induction motors and BLDC motor                                                                | 05 | CO4 |
| V   | 05 | Special Purpose Diodes: (Numerical are not expected) Characteristics and operation of Zener Diode and application as a voltage regulator. Basic and structure of LED. Application of LED in indicative and lighting displays.                                                           | 04 | CO5 |
| VI  | 06 | Introduction to Transistors: (Numerical are not expected) structure and operation of BJT. BJT configurations (only common emitter). FET structure and operation. Application of BJT and FET in amplification, switching and oscillators.                                                | 04 | CO6 |

#### **Text Books:**

1. V. N. Mittal and Arvind Mittal "Basic Electrical Engineering" Tata McGraw Hill, (Revised Edition)
2. Vincent Del Toro "Electrical Engineering Fundamentals", PHI Second edition, 2011
3. Edward Hughes "Hughes Electrical and Electronic Technology", Pearson Education (Tenth edition)
4. D P Kothari and I J Nagrath "Theory and Problems of Basic Electrical Engineering", PHI 13th edition 2011.
5. M. Naidu, S. Kamakshiah "Introduction to Electrical Engineering" McGraw-Hill Education, 2004.
6. B.R Patil "Basic Electrical Engineering" Oxford Higher Education,
7. Electronic Devices and Circuit Theory" by Robert L. Boylestad and Louis Nashelsky

#### **References:**

1. B.L. Theraja "Electrical Engineering " Vol-I and II
2. S.N. Singh, "Basic Electrical Engineering" PHI , 2011 Book

#### **Assessment:**

##### **Internal Assessment Test (IAT) for 20 marks each:**

- IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

#### **End Semester Theory Examination:**

➤ **Question paper format**

- Question Paper will comprise a total of **six questions each carrying 15 marks** Q.1 will be **compulsory** and should **cover the maximum contents of the syllabus**
- **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered

| Course Code | Course Name                                      | Teaching Scheme (Contact Hours) |        |      | Credits Assigned |        |      |       |
|-------------|--------------------------------------------------|---------------------------------|--------|------|------------------|--------|------|-------|
|             |                                                  | Theory                          | Pract. | Tut. | Theory           | Pract. | Tut. | Total |
| ESL102      | Basic Electrical and Electronics Engineering Lab | --                              | 2      | -    | --               | 1      | -    | 1     |

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|-------------|--------------------------------------------------|--------------------------------|--------|------------------------|--------------|------------------------|-----------|--------------|-------|
|             |                                                  | Internal Assessment Test (IAT) |        |                        | End Sem Exam | Exam Duration (in Hrs) |           |              |       |
|             |                                                  | IAT-I                          | IAT-II | IAT-I + IAT-II (Total) |              |                        |           |              |       |
| ESL102      | Basic Electrical and Electronics Engineering Lab | --                             | --     | --                     | --           | --                     | 25        | 25           | 50    |

**Lab Objectives:**

1. To impart the basic concept of network analysis and its application.
2. To provide the basic concept of AC circuit analysis and its application.
3. To illustrate the operation of the transformer.
4. To illustrate the operation of machines.
5. To explain the Zener diode voltage regulation characteristic.
6. To explain the BJT and FET as switches and amplifiers.

**Lab Outcomes:**

- 1) Interpret and analyze the behavior of DC circuits using network theorems.
- 2) Perform and infer experiments on single-phase and three-phase AC circuits
- 3) Illustrate the performance of a single-phase transformer
- 4) Illustrate the performance of A.C. machine and DC Motor
- 5) Perform an experiment on voltage regulation characteristics of Special diode
- 6) Perform an experiment on the VI characteristic Transistor.

**List of Experiments.**

| Sr No | List of Experiments                                                                                                                                                     | Hrs | LO Mapping |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------|
| 01    | Basic safety precautions. Introduction and use of measuring instruments - voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors, and inductors | 01  | LO1        |
| 02    | To measure output voltage across load resistor/current through load resistor and verify the result using Mesh and Nodal analysis                                        | 01  | LO1        |
| 03    | Verification of Superposition Theorem.                                                                                                                                  | 02  | LO1        |
| 04    | Verification Thevenin's and Norton's theorem                                                                                                                            | 02  | LO1        |
| 05    | Verification Maximum Power Transfer Theorem.                                                                                                                            | 02  | LO1        |
| 06    | To find the resistance and inductance of a coil connected in series with a pure resistance using the voltmeter method                                                   | 02  | LO2        |
| 07    | To measure the relationship between phase and line, currents and voltages in three-phase system (star & delta)                                                          | 02  | LO2        |
| 08    | To demonstrate cut-out sections of the single-phase transformer.                                                                                                        | 02  | LO3        |
| 09    | To demonstrate cut-out sections of the DC machine                                                                                                                       | 02  | LO4        |
| 10    | To plot Zener diode voltage regulation characteristics                                                                                                                  | 02  | LO5        |
| 11    | To demonstrate the application of LED in indicative and lighting display                                                                                                | 02  | LO5        |
| 12    | To demonstrate the application of BJT as a switch                                                                                                                       | 02  | LO5        |
| 13    | To demonstrate BJT/FET as an amplifier                                                                                                                                  | 02  | LO6        |

| Sr No | List of Assignments / Tutorials                                              | Hrs | LO Mapping |
|-------|------------------------------------------------------------------------------|-----|------------|
| 01    | Assignment on Basic electrical safety practices                              | 02  | LO1        |
| 02    | Numerical assignment on Mesh analysis and nodal analysis                     |     | LO1        |
| 03    | Numerical assignment on Thevenin, Norton, and maximum power transfer theorem |     | LO1        |
| 04    | Numerical assignment on series and parallel circuits                         |     | LO2        |
| 05    | Assignment on single-phase transformer                                       |     | LO2        |
| 06    | Assignment on DC and AC machine                                              |     | LO4        |
| 07    | Assignment on special purpose diodes                                         |     | LO5        |
| 08    | Assignment on BJT and FET                                                    |     | LO6        |

**Online Resources:**

| Sr. No. | Website Name                                                                                           |
|---------|--------------------------------------------------------------------------------------------------------|
| 5.      | All About Circuits ( <a href="https://www.allaboutcircuits.com">https://www.allaboutcircuits.com</a> ) |

|    |                                                                                     |
|----|-------------------------------------------------------------------------------------|
| 6. | Circuit Lab ( <a href="https://www.circuitlab.com">https://www.circuitlab.com</a> ) |
| 3. | Tinkercad ( <a href="https://www.tinkercad.com">https://www.tinkercad.com</a> )     |

**Assessment:**

**Term Work:** Term Work shall consist of at least 08 to 10 practicals based on the above list. Also, Term work Journal must include at least 6 assignments.

**Term Work Marks:** 25 Marks (Total marks) = 15 Marks (Experiment) + 5 Marks (Assignments) + 5 Marks (Attendance)

**Practical& Oral Exam:** An Oral & Practical exam will be held based on the above syllabus.

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|-------------|---------------------------------------|------------------------------------|--------|------|------------------|--------|------|-------|
|             |                                       | Theory                             | Pract. | Tut. | Theory           | Pract. | Tut. | Total |
| AEC101      | Professional Communication and Ethics | 02                                 | -      | -    | 02               | -      | -    | 02    |

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|-------------|---------------------------------------|--------------------------------|--------|------------------------|--------------|------------------------|-----------|--------------|-------|
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|             |                                       | IAT-I                          | IAT-II | IAT-I + IAT-II (Total) |              |                        |           |              |       |
| AEC101      | Professional Communication and Ethics | 15                             | 15     | 30                     | 45           | 1.5                    | --        | --           | 75    |

**Rationale**

This course has been designed to hone the communicative abilities of First Year Engineering students by providing them skill-based training on LSRW (Listening-Speaking-Reading-Writing) to prepare